

FIX ANY PC PROBLEM YOURSELF

BEGINNER EDITION

Save \$100+ every time your computer breaks. No tech skills needed.

8 COMPLETE REPAIR SECTIONS • PRINTABLE CHECKLISTS • PLAIN ENGLISH GLOSSARY

Written for home users, parents, and small business owners

Fix Any PC Problem Yourself — Beginner Edition

First Edition

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Disclaimer: This guide is intended for educational and informational purposes. The troubleshooting steps described are generally safe and widely practiced. However, opening computer hardware carries inherent risks. The author assumes no liability for damage to hardware, loss of data, or any other harm arising from following the steps in this guide. When in doubt, consult a qualified technician.

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Introduction

85% of PC problems have a simple fix	\$150+ average repair shop visit cost	8 problems covered in this guide	0 tech skills required
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Your computer stopped working. Maybe it won't start. Maybe the screen is black. Maybe it's so slow you want to throw it out the window. Whatever happened — you're in the right place.

This guide was written for one specific reason: **most computer problems that cost people \$100–\$300 at a repair shop can be fixed at home in under an hour.** Not because computers are simple — but because the most common failures have predictable causes and straightforward fixes.

What Makes This Guide Different

- **Eight complete repair sections** — covering the problems most people actually face, not obscure edge cases
- **Step-by-step instructions written in plain English** — no assumed knowledge, no jargon without explanation
- **Printable checklists** at the end of each section so you can work through fixes without staring at a screen
- **A complete pre-shop checklist** — so if you do need a repair shop, you go in prepared and don't overpay
- **A full glossary** of every technical term used, explained in one or two plain sentences
- **Clear stopping points** — this guide never asks you to attempt something risky or beyond a beginner's ability

How to Use This Guide

You don't have to read this front to back. Go to the Quick Diagnosis page (page 6), find your problem, and jump straight to that section. Each section is self-contained — everything you need

is right there.

Print the checklist at the end of each section before you start. Working from paper while troubleshooting is much easier than switching between a guide and your broken computer.

Before You Start

Read this page once before you begin any section. It covers the tools you might need, the safety rules that protect you and your computer, and what to realistically expect.

Tools You Might Need

Item	What It's Used For	Required?
Phillips screwdriver (small)	Opening the computer case to reseal RAM or check cables	Sometimes
USB flash drive (any size)	Creating a Windows recovery or reinstall tool	Section H only
Spare monitor or TV with HDMI	Testing whether your screen is the problem	Optional
Can of compressed air	Cleaning dust from fans and vents (Section D)	Section D only
A second working computer or phone	Searching online while your PC is being fixed	Helpful

Safety Rules — Please Read All of These

These five rules apply every time you open or touch the inside of a computer. They are simple and non-negotiable.

- **Power off completely.** Hold the power button for 5 full seconds until all lights go off. Sleep mode is not off.
- **Unplug the power cable from the wall.** Do this before opening the case or touching anything inside — even if the computer appears to be off.
- **Ground yourself before touching components.** Touch the metal frame of the computer case with your bare hand before touching any internal part. This discharges static electricity that can silently destroy components.

- **Never force anything.** Every cable, RAM stick, and connector has a specific way it fits. If it isn't sliding in easily, stop and re-check the orientation.
- **Take photos before disconnecting anything.** Use your phone to photograph cable positions before unplugging them. This makes reconnecting much easier.

If you smell burning, see smoke, or hear a loud pop from inside your computer — unplug it immediately and do not attempt any further troubleshooting yourself. This indicates a component failure that requires professional assessment.

Quick Diagnosis

Find Your Problem in 60 Seconds

Read each description below. Find the one that best matches your situation. Go directly to that section — you don't need to read anything else first.

What's happening?	Go to
Nothing happens when you press the power button. No lights, no fans, no sounds.	Section A Page 7
The computer turns on (you hear fans or see lights) but the screen is completely black.	Section B Page 10
The computer works but everything is painfully slow. Programs take forever to load.	Section C Page 13
The computer shuts off by itself, restarts randomly, or the fan is extremely loud.	Section D Page 17
You see a blue screen with white text and an error code or frowning face.	Section E Page 20
You see 'No boot device found' or 'Operating system not found' on screen.	Section F Page 24
Wi-Fi shows connected but pages won't load, or you can't connect at all.	Section G Page 27
Windows gets stuck loading, loops, or shows 'Automatic Repair' repeatedly.	Section H Page 31

Not sure which section applies? Your symptoms may overlap. Start with whichever description feels closest — the opening of each section describes the problem in more detail so you can confirm you're in the right place.

Is This Your Problem?

You press the power button and absolutely nothing happens. The fans don't spin. No lights flicker. No sound. The computer behaves as if it's not receiving any power at all.

This is the most common PC problem and also one of the most frequently misdiagnosed. Before assuming the worst, nearly 60% of 'dead computer' cases are caused by a loose cable, a dead outlet, or a tripped power strip.

Understanding What Could Be Wrong

When a computer gets no power at all, there are four possible culprits — and we're going to check the simplest ones first:

- **The wall outlet or power strip** — not delivering power
- **The power cable** — loose or damaged
- **The power supply switch** — flipped to OFF
- **The RAM** — loose in its slot, preventing startup
- **The power supply unit (PSU)** — failed internally (less common, needs a technician)

Step-by-Step Fix

Work through these steps in order. After each step, plug in and try the power button. Stop as soon as it works.

1

Test your wall outlet. Plug a lamp or phone charger into the same outlet. If that device doesn't work, the outlet is dead — try a different one. If it does work, continue to Step 2.

2

Check your power strip or surge protector. If your computer plugs into a power strip, make sure it's switched ON. Look for a reset button (usually a small red button) and press it once. Try the computer again.

3	Reseat the power cable at the back of the computer. Find the thick cable at the back of your computer (where it plugs into the machine itself). Pull it out fully and push it firmly back in. It should feel solid, not wobbly.
4	Check the power supply switch. On the back of the computer, next to where the power cable plugs in, there is often a small rocker switch. Make sure it shows the symbol 'I' or 'I' (ON), not 'O' (OFF). Toggle it if needed.
5	Perform a power drain. Unplug the power cable from the wall. Hold the power button down for a full 30 seconds. This discharges any residual electricity inside the system. Plug back in and try again.
6	Reseat the RAM modules. RAM that has worked loose will prevent the computer from starting completely. Follow the detailed RAM reseating instructions below.

How to Reseat RAM — Detailed Instructions

RAM (Random Access Memory) is a flat stick — usually green or black, about 5 inches long — that plugs into the main board. Your computer has one or two. This is the safest internal repair a beginner can make.

1	Power off and unplug the power cable from the wall.
2	Open the side panel of the case. Most have two screws on the back-left edge, or a thumb-latch you slide to the side.
3	Touch the metal frame of the case with your bare hand before touching anything else inside.
4	Locate the RAM sticks. They are long flat sticks plugged into slots on the main board. There may be 1, 2, or 4 of them.
5	At each end of a RAM stick, you'll see a small plastic clip. Gently press both clips outward at the same time — the stick pops up slightly.

6	Remove the RAM stick completely and set it face-up on a clean, dry surface.
7	Wait 15 seconds. Then line up the RAM stick with the slot (there is a notch in the middle — it only fits one way) and press it firmly down until both clips snap back into place.
8	If you have two RAM sticks, try booting with just one installed. Then swap to the other. This identifies if one stick is faulty.
9	Close the case, plug in, and test.

If this worked: Your computer powers on normally. If RAM reseating fixed it, note this — if it happens again, that RAM stick may need to be replaced. RAM prices vary significantly — budget \$40–\$120+ depending on your system's spec.

If this didn't work: If none of these steps produced any response, the power supply unit (PSU) has likely failed. See the 'When to Stop' guidance at the end of each section, or go directly to the pre-shop checklist on page 35.

Section A — Troubleshooting Checklist	Done?
Tested the wall outlet with another device	☐
Checked and reset the power strip / surge protector	☐
Firmly reseated the power cable at the back of the computer	☐
Confirmed the power supply switch is set to ON (I)	☐
Performed a 30-second power drain	☐
Reseated all RAM sticks	☐
Tried booting with one RAM stick at a time (if you have two)	☐

Is This Your Problem?

Your computer turns on — the fans spin, lights may come on — but the monitor shows nothing. Completely black. You might hear Windows start-up sounds, or you might hear nothing but the fans running.

This symptom has three distinct causes that need to be separated before you can fix anything:

- **Monitor/cable problem:** The computer is working fine — the image just isn't reaching your screen
- **RAM or hardware problem:** The computer is failing before it can output anything to the screen
- **Software/driver problem:** Windows starts to load but crashes before showing the desktop

Phase 1 — Check the Monitor and Cables

1

Check the monitor's own power. Look for a power button on the monitor itself — usually on the bottom-front edge or underside. The monitor has its own power, separate from the computer. Press it. Look for a small indicator light (blue or white = on, amber = standby, off = unpowered).

2

Reseat the video cable. Unplug the cable going from the back of your monitor to the back of your computer. Plug it firmly back in at both ends. If the cable has screws, tighten them finger-tight.

3

Confirm you're plugged into the right port. If your computer has a separate graphics card (a card with its own fan, usually near the bottom of the back panel), make sure your monitor cable plugs into THAT card — not into the ports higher up on the back panel. Using the wrong port is extremely common.

4

Try a different cable type. If you're using HDMI, try DisplayPort. If you have a VGA adapter, try that. Cables fail more often than people realize.

5

Test with a different monitor or TV. Plug your computer into a TV using an HDMI cable. If the desktop appears on the TV, your original monitor or its cable is the problem — not the computer.

Phase 2 — Check Internal Hardware

If Phase 1 didn't help — your monitor is fine and the cable is fine — the issue is inside the computer.

1

Reseat the RAM sticks. Follow the RAM reseating steps from Section A. A bad RAM connection is one of the most common causes of a black screen at startup.

2

Listen for beep codes. When a computer can't start properly, it may beep in a pattern. Count the beeps carefully (e.g., 3 short, 1 long). Write down the pattern, then search online: '[your computer brand] beep code [pattern]'. This can identify the exact failed component.

3

Reseat the graphics card (if you have one). Power off and unplug. The graphics card is in a long slot near the bottom of the main board. There's a clip at one end holding it in. Release the clip, remove the card, and firmly push it back in until it clicks. Reconnect any power cables going to the card.

4

Test with the graphics card removed (if applicable). Some computers have basic video output built into the main board. Remove the graphics card and plug your monitor into the built-in port. If you get a picture, the graphics card has failed.

Phase 3 — Software or Driver Issue

If the computer was working and then a Windows update or driver install caused the black screen, this is a software issue. Go to Section H (Windows Won't Boot — Recovery Guide) on page 31.

If this worked: The screen comes on and you can see your desktop. If a cable or port swap fixed it, note which port/cable works. Consider replacing the faulty cable.

If this didn't work: If reseating RAM and the graphics card didn't help, and the monitor tests fine on another device, the graphics card or motherboard has likely failed. See the pre-shop checklist on page 35.

Section B — Troubleshooting Checklist	Done?
Confirmed the monitor has its own power and the power button was pressed	☐
Reseated the video cable at both the monitor and computer ends	☐
Confirmed monitor is plugged into the graphics card port (not the motherboard port)	☐
Tested with a different cable type (HDMI / DisplayPort / VGA)	☐
Tested with a different monitor or TV	☐
Reseated all RAM sticks	☐
Listened for beep codes and noted the pattern	☐
Reseated the graphics card (if installed)	☐
Tested with graphics card removed (if applicable)	☐

Is This Your Problem?

Your computer starts and you can use it — but everything crawls. Programs take minutes to open. Switching between windows feels like wading through mud. The computer might feel 'stuck' even when you're not doing anything.

Common causes of slowness, ranked by frequency:

- Too many programs starting automatically at boot
- Storage drive is more than 85% full
- Too many background processes running
- Malware or adware consuming resources
- A failing or fragmented hard drive
- Insufficient RAM for modern software
- Overheating causing the CPU to throttle itself (see also Section D)

Fix 1 — Disable Startup Programs

This is the single most impactful fix for most slow computers and takes under five minutes. Every installed application tends to add itself to your startup list. Over years, this list grows until your computer takes 5+ minutes just to become usable after turning on.

1

Press **Ctrl + Alt + Delete** simultaneously. Choose **Task Manager** from the menu that appears.

2

In Task Manager, click the **Startup** tab at the top. (If you see a simplified view, click 'More details' at the bottom first.)

3

You'll see a list of programs with a 'Status' column. 'Enabled' means it runs every time you start your computer.

4	Look at the 'Startup impact' column — it will say Low, Medium, or High. Focus on High-impact items first.
5	Click on any program you don't need running at startup. Click the Disable button in the bottom-right.
6	Safe to disable: Spotify, Discord, Steam, Zoom, Teams, Skype, OneDrive, Slack, iTunes, Adobe Updater, any gaming launcher.
7	Do NOT disable: Windows Defender, your antivirus, audio drivers, or anything labelled 'Windows'.
8	Restart your computer when done and measure the difference.

Fix 2 — Free Up Storage Space

When a storage drive gets too full, Windows slows dramatically — it needs free space to operate as virtual memory and for temporary files. Keep your drive below 80% full as a rule.

1	Press the Windows key and type This PC , then press Enter.
2	Look at the bar under 'Local Disk (C:)'. If it's red or orange, you need to free space urgently.
3	Press the Windows key and type Disk Cleanup . Press Enter. Select your C: drive and click OK.
4	Windows will calculate recoverable space. Check all boxes in the list — all items are safe to delete.
5	Click Clean up system files as well. This typically finds several gigabytes more.
6	After Disk Cleanup finishes: right-click the Recycle Bin on your desktop and choose 'Empty Recycle Bin'.
7	Open File Explorer and go to your Downloads folder. Delete anything you no longer need.

8

Press the Windows key and type **Add or Remove Programs**. Uninstall any software you haven't used in months.

Fix 3 — Check for Malware

Malware and adware are a frequent cause of sudden slowness — they run constantly in the background consuming your CPU and memory.

1

Press the Windows key and type **Windows Security**. Press Enter.

2

Click **Virus & threat protection**.

3

Click **Quick scan**. Wait for it to complete — this takes 5–15 minutes.

4

If threats are found, follow the on-screen prompts to remove them.

5

For a deeper scan: click 'Scan options' and select 'Full scan'. This can take an hour but is thorough.

6

Restart your computer after the scan completes, even if no threats were found.

Fix 4 — Check What's Consuming Resources Right Now

1

Open Task Manager (Ctrl + Alt + Delete → Task Manager).

2

Click the **Processes** tab. Click the **CPU** column header to sort by CPU usage — highest first.

3

Look for any process using more than 20–30% CPU consistently. Note its name.

4

Click the **Memory** column header and check for anything using excessive RAM.

5

If you see an unfamiliar process using high resources: right-click it, choose 'Search online', and look it up before doing anything.

6

Legitimate Windows processes you can ignore: System, svchost.exe, Windows Defender, antivirus processes.

7

If a process is clearly a rogue program, right-click it and choose 'End task'. Then uninstall the associated program.

Fix 5 — Upgrade to an SSD (The Biggest Improvement)

If your computer still has a traditional spinning hard drive (HDD), upgrading to an SSD is the single most transformative improvement available. Boot times go from minutes to seconds. Programs open instantly.

To check your drive type: Press Windows + R, type **dfrgui**, press Enter. In 'Optimize Drives', look at the 'Media type' column. 'Hard disk drive' = traditional. 'Solid state drive' = already an SSD. If you have a hard disk drive, an SSD upgrade will transform your computer.

SSD Upgrade Overview

Factor	Old Hard Drive (HDD)	SSD Upgrade
Boot time	2–5 minutes	8–12 seconds
Program open speed	Slow	Near-instant
Noise	Audible spinning/clicking	Completely silent
Cost	Already installed	\$40–\$80 for the drive
DIY difficulty	—	Moderate (or have a shop do it)

If this worked: Measurable improvement in speed. Keep storage below 80% and run Disk Cleanup monthly.

If this didn't work: If the computer is still very slow after all fixes, it may be aging hardware or a failing drive. Run a drive health check (Fix 6 below) and consult the pre-shop checklist.

Section C — Troubleshooting Checklist

Done?

Disabled unnecessary startup programs in Task Manager	■
Ran Disk Cleanup (including 'Clean up system files')	■
Emptied the Recycle Bin	■
Deleted unneeded files from Downloads folder	■
Uninstalled unused programs	■
Ran a Windows Security full scan	■
Checked Task Manager Processes tab for resource hogs	■
Checked drive type (HDD vs SSD) — considered upgrade if HDD	■

Is This Your Problem?

Your computer shuts off by itself without warning, restarts on its own, or the fan runs at full speed constantly (very loud). The computer may slow down significantly under load before shutting off. It might also feel very hot to the touch.

Overheating is serious. Repeated thermal shutdowns can permanently damage your CPU, GPU, or motherboard. Address this promptly.

Why Computers Overheat

Computers generate significant heat during operation. Fans move cool air in and hot air out. When this system fails — due to dust, a dead fan, or dried-out thermal paste — temperatures spike and the computer shuts down as a safety measure to protect itself.

- **Dust buildup** — the most common cause; clogs vents and smothers heatsinks
- **Fan failure** — one or more fans have stopped spinning
- **Dried thermal paste** — the compound between the CPU and heatsink has failed (common in computers 4+ years old)
- **Blocked vents** — computer placed on carpet, against a wall, or in an enclosed space
- **Failing power supply** — can cause erratic shutdowns that mimic overheating

Fix 1 — Clean the Dust (Do This First)

- | | |
|---|--|
| 1 | Power off the computer completely and unplug from the wall. |
| 2 | Take the computer to a well-ventilated area — outside if possible, or near an open window. |
| 3 | Open the side panel of the case. |

4	Use a can of compressed air to blow dust out of all fans, heatsinks (the finned metal structures), and vents. Hold the can upright and use short bursts.
5	Hold fans still with a finger while blowing them — spinning a fan with compressed air can damage the bearing.
6	Pay special attention to the CPU heatsink (the large finned structure with a fan on top of it) and any GPU fans.
7	Also blow out the power supply vents at the back or top of the case — dust accumulates there too.
8	Use a dry cloth or cotton swab to wipe visible dust off surfaces.
9	Replace the side panel, plug in, and test.

Fix 2 — Check Fan Operation

1	Power on the computer and immediately look inside (with the side panel open) to verify all fans are spinning.
2	The CPU fan (on top of the large heatsink), GPU fan (on the graphics card), and case fans should all be spinning within a few seconds of power-on.
3	If any fan is not spinning, it has failed. Case fans are relatively affordable (\$10–\$30) and straightforward to replace. CPU cooler fans vary more in price. GPU fan replacement is more complex and may require a full card replacement.
4	You can also download a free tool called HWMonitor (by CPUID) to check temperatures. CPU temperature at idle should be below 50°C. Under load, below 80°C.

Fix 3 — Improve Airflow

1	Make sure the computer is not on carpet (blocks the bottom intake vents). Place it on a hard flat surface.
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2	Ensure at least 6 inches of clearance on all sides — especially the back and top where hot air exits.
3	If the computer is inside a cabinet or enclosed desk, move it to an open area.
4	Check that no cables inside the case are blocking airflow paths between the front and back.

Fix 4 — Thermal Paste (Advanced, for computers 4+ years old)

Thermal paste sits between the CPU and its heatsink and conducts heat efficiently. After 4–7 years it dries out and loses effectiveness, causing temperatures to rise significantly. Replacing it can drop temperatures by 15–25°C.

Thermal paste replacement requires removing the CPU cooler and is moderately advanced. If you're not comfortable with this, any repair shop can do it for \$20–\$40 in labor — it's quick for a technician.

If this worked: The computer runs cooler, the fan noise reduces, and random shutdowns stop. Make dust cleaning every 6 months a permanent habit.

If this didn't work: If cleaning and improving airflow didn't help and fans are working, the thermal paste likely needs replacing, or the power supply may be failing. See page 35.

Section D — Troubleshooting Checklist	Done?
Cleaned all dust from fans, heatsinks, and vents with compressed air	☐
Verified all fans (CPU, GPU, case fans) are spinning on power-on	☐
Moved computer off carpet to a hard surface with clear airflow	☐
Ensured at least 6 inches of clearance around the computer	☐
Checked temperatures with HWMonitor (CPU idle below 50°C)	☐
Considered thermal paste replacement (if computer is 4+ years old)	☐

Is This Your Problem?

Your computer suddenly shows a blue (or occasionally green) screen with white text and either an error code, a percentage bar, or a frowning face emoji. The computer then restarts automatically.

A single BSOD is not necessarily a crisis — Windows can crash from a one-time software glitch. The concern is repeated BSODs, or a BSOD that prevents the computer from ever fully starting.

Reading the Error Code

The most useful thing on a BSOD screen is the error code — a phrase in ALL_CAPS with underscores, like **MEMORY_MANAGEMENT** or **DRIVER_IRQL_NOT_LESS_OR_EQUAL**. This tells you what failed.

Take a photo of the BSOD screen with your phone before the computer restarts. The error code is your primary diagnostic tool.

Error Code	Likely Cause	First Step
MEMORY_MANAGEMENT	Faulty or failing RAM	Run Windows Memory Diagnostic
DRIVER_IRQL_NOT_LESS_OR_EQUAL	Bad driver (often after an update)	Roll back recent driver updates
NTFS_FILE_SYSTEM	Hard drive error or corruption	Run CHKDSK (Fix 3 below)
CRITICAL_PROCESS_DIED	Corrupted Windows system files	Run SFC scan (Fix 4 below)
IRQL_NOT_LESS_OR_EQUAL	Driver conflict or RAM issue	Update drivers / test RAM
KERNEL_SECURITY_CHECK_FAILURE	Driver or memory corruption	Run SFC and memory test

PAGE_FAULT_IN_NONPAGED_AREA	RAM, driver, or storage issue	Run memory diagnostic
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Fix 1 — Check for Recent Changes

1	Think back: did the BSOD start after a Windows update, a new driver installation, or a newly installed program?
2	If yes — that update or installation is likely the cause. Uninstalling it is your first move.
3	To uninstall a recent Windows update: Settings → Windows Update → Update History → Uninstall Updates.
4	To roll back a driver: Device Manager → find the device → right-click → Properties → Driver tab → 'Roll Back Driver'.

Fix 2 — Test Your RAM

1	Press the Windows key and type Windows Memory Diagnostic . Press Enter.
2	Choose 'Restart now and check for problems'.
3	Your computer will restart and run the memory test — this takes 10–20 minutes.
4	When done, Windows restarts and shows you the results. If errors are found, a RAM stick is failing and needs to be replaced.
5	Also try reseating the RAM sticks (instructions in Section A) even if no errors show — physical connection issues can cause BSODs without triggering test errors.

Fix 3 — Check the Drive for Errors (CHKDSK)

1	Press the Windows key and type Command Prompt . Right-click it and choose 'Run as administrator'.
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2	In the black window that appears, type exactly: chkdsk C: /f /r and press Enter.
3	You'll be asked if you want to schedule this for the next restart. Type Y and press Enter.
4	Restart your computer. Windows will run the disk check before loading — this can take 30–90 minutes depending on drive size.
5	Any errors found will be repaired automatically. Review the results when Windows loads.

Fix 4 — Repair Windows System Files (SFC Scan)

1	Press the Windows key and type Command Prompt . Right-click → 'Run as administrator'.
2	Type: sfc /scannow and press Enter.
3	Wait for the scan to complete — it takes 5–20 minutes and shows a progress bar.
4	If it reports 'Windows Resource Protection found corrupt files and repaired them' — restart and test.
5	If BSODs continue after repairing system files, go to Section H (Windows Recovery) on page 31.

If this worked: BSODs stop occurring after fixing the identified issue. Monitor for recurrence over the next 48 hours. If it stays stable, you're done.

If this didn't work: If BSODs continue after all fixes, especially if they happen during or before Windows loads, go to Section H (Windows Won't Boot — Recovery Guide) or consult the pre-shop checklist.

Section E — Troubleshooting Checklist	Done?
Photographed the BSOD error code	☐
Looked up the error code in the table above	☐

Checked for and rolled back any recent Windows updates or driver installs	■
Ran Windows Memory Diagnostic — noted results	■
Reseated all RAM sticks	■
Ran CHKDSK C: /f /r and restarted	■
Ran SFC /scannow and restarted	■

Is This Your Problem?

You turn on your computer and instead of Windows loading, you see an error on a black screen. Common messages include:

- **"No boot device found"**
- **"Operating system not found"**
- **"Reboot and select proper Boot device"**
- **"NTLDR is missing"**
- **"Boot device inaccessible"**

This means your computer cannot find the drive where Windows is installed. The drive is either disconnected, not being detected, or has failed.

What Is a SATA Cable?

A SATA cable is the thin flat cable (usually red or black) that connects your hard drive or SSD to the main board. A separate, wider cable provides power to the drive. Both need to be firmly connected. Loose cables are the #1 cause of 'no boot device' errors and take 2 minutes to fix.

Fix 1 — Check and Reseat All Drive Cables

1

Power off the computer and unplug from the wall. Touch the metal case frame before touching anything inside.

2

Locate your drive. It's a flat rectangular box — 2.5 inch (laptop-style) or 3.5 inch (desktop). Two cables connect to it.

3

Unplug the SATA data cable (thin, flat, L-shaped connector) from both the drive and the main board. Reconnect both ends firmly.

4	Unplug and reconnect the power cable (wider, with multiple colored wires). Make sure it's fully seated.
5	If there are multiple SATA ports on the main board, try moving the data cable to a different port. Some ports fail while others work.
6	Plug in and restart. If Windows loads normally, you're done.

Fix 2 — Check the BIOS

The BIOS is the low-level settings screen that runs before Windows loads. It lists every piece of hardware connected to your computer. If the BIOS can see your drive, it's physically alive — the problem is Windows, not the drive.

1	Turn on the computer. Immediately start pressing the BIOS key repeatedly until the BIOS opens. Keys by brand: Dell = F2, HP = F10 or Esc, Lenovo = F1 or F2, ASUS / MSI / Gigabyte = Delete key.
2	If you miss the window, the computer will just boot normally (or fail to). Restart and try again.
3	Once in BIOS, navigate using arrow keys. Look for a section called 'Boot', 'Storage', or 'SATA Configuration'.
4	Check if your drive name appears in the list. It might show a brand name (Seagate, Western Digital, Samsung, Crucial) and a capacity (e.g., '500GB').
5	If the drive IS listed: The hardware is fine. The issue is Windows — go to Section H (Windows Recovery, page 31).
6	If the drive is NOT listed: The computer cannot detect the drive at any level. Continue to Fix 3.
7	Exit BIOS without saving changes: press Escape, then select 'Exit Without Saving'.

Fix 3 — Diagnosing a Failed Drive

If the drive is missing from the BIOS after you've reseated all cables — especially if you hear clicking, grinding, or repetitive beeping from inside the computer — the drive has likely failed mechanically. Stop attempting to restart it. Each restart can worsen the damage and reduce the chance of professional data recovery.

Signs of a failing or failed drive:

- Loud clicking or grinding noises from inside the computer
- Drive disappears from BIOS on every restart
- Drive becomes extremely hot to the touch
- Files randomly disappear or refuse to open
- Windows takes extremely long to access files or copy data
- CHKDSK reports large numbers of bad sectors

If your drive has failed, a professional data recovery service may still be able to retrieve your files — but success rates decrease with every additional power cycle. If your files are important, stop and consult a professional before doing anything else.

If this worked: The error disappears and Windows loads normally. Back up your files immediately — a drive that lost its cable connection can be a sign of early drive failure.

If this didn't work: Drive not detected in BIOS and cables are confirmed good. The drive has likely failed. If you need the data, contact a data recovery specialist. For the computer itself, a replacement drive + Windows reinstall is the path forward.

Section F — Troubleshooting Checklist	Done?
Firmly reseated the SATA data cable at both the drive and the main board	☐
Firmly reseated the power cable at the drive	☐
Tried a different SATA port on the main board	☐
Entered BIOS and checked whether the drive appears in the device list	☐

Listened for clicking or grinding sounds (stop if heard)



Is This Your Problem?

Wi-Fi shows as connected but pages don't load. Or the Wi-Fi icon shows no connection at all. Or other devices on your network work fine but your computer can't get online.

Diagnose which situation you're in:

What you see	Likely cause and where to go
No Wi-Fi networks appear at all	Driver issue or Wi-Fi adapter problem → Fix 3 or Fix 4
Networks appear but won't connect	Password/authentication issue or router problem → Fix 1
Shows 'Connected' but no internet	IP address or DNS issue → Fix 2
Other devices work but not this computer	Issue is with the computer specifically → Fix 2 or Fix 3
No devices can connect	Router/modem issue — Fix 1 first

Fix 1 — Restart Everything (Do This First)

1	Turn off your computer completely.
2	Unplug your router (the box with the Wi-Fi signal) from the wall. Wait 30 seconds.
3	If you have a separate modem (a box that connects to a cable or phone line), unplug that too.
4	Plug the modem back in first (if separate). Wait 60 seconds for it to fully reconnect.
5	Plug the router back in. Wait another 60 seconds.
6	Turn your computer back on and try connecting.

- | | |
|---|--|
| 7 | This fixes roughly 40% of Wi-Fi problems on its own. |
|---|--|

Fix 2 — Reset Network Settings on the Computer

- | | |
|---|---|
| 1 | Press the Windows key and type Command Prompt . Right-click → 'Run as administrator'. |
| 2 | Type each of the following commands, pressing Enter after each one. Wait for each to complete before typing the next: |
| 3 | Type: netsh winsock reset — then Enter |
| 4 | Type: netsh int ip reset — then Enter |
| 5 | Type: ipconfig /release — then Enter |
| 6 | Type: ipconfig /flushdns — then Enter |
| 7 | Type: ipconfig /renew — then Enter |
| 8 | Close the Command Prompt window and restart your computer. |

Fix 3 — Update or Reinstall the Wi-Fi Driver

- | | |
|---|---|
| 1 | Right-click the Start button and choose Device Manager . |
| 2 | Expand 'Network Adapters' by clicking the arrow next to it. |
| 3 | Find your Wi-Fi adapter (often has 'Wireless' or 'Wi-Fi' in the name). Right-click it. |
| 4 | Choose Update driver → 'Search automatically for drivers'. Let Windows look for an update. |
| 5 | If no update is found, try: right-click the adapter → Uninstall device . Confirm. |

6	Restart your computer. Windows will automatically reinstall the driver on startup.
7	If the adapter doesn't appear in Device Manager at all, see Fix 4.

Fix 4 — Check Physical Wi-Fi Switch / Settings

1	Some laptops have a physical switch on the side or front that toggles Wi-Fi on and off. Check that it's in the ON position.
2	Some laptops use a keyboard shortcut (often Fn + F2, F3, or a key with a Wi-Fi symbol). Press it to toggle Wi-Fi.
3	Press the Windows key and type Network troubleshooter . Run the built-in troubleshooter — it often automatically fixes common issues.
4	Check if Airplane Mode is on: look for the airplane icon in the system tray (bottom right). Click it and make sure Airplane Mode is off.
5	Settings → Network & Internet → Wi-Fi. Make sure Wi-Fi is toggled ON.

Fix 5 — Change DNS Server

Sometimes the issue is with your internet provider's DNS server — the system that translates website names into addresses. Switching to a public DNS server can instantly fix 'connected but no internet' issues.

1	Press the Windows key and type Network Connections . Press Enter.
2	Right-click your Wi-Fi adapter and choose Properties .
3	Double-click Internet Protocol Version 4 (TCP/IPv4) .
4	Select 'Use the following DNS server addresses'.

5

Enter: Preferred DNS: **8.8.8.8** — Alternate DNS: **8.8.4.4** (these are Google's public DNS servers).

6

Click OK, then close all windows. Try loading a webpage.

If this worked: Internet access is restored. If restarting the router fixed it, consider setting up a weekly router restart schedule — it keeps connections stable.

If this didn't work: If none of these fixes restored connectivity and other devices work on the same Wi-Fi, the Wi-Fi adapter in the computer may have failed. An external USB Wi-Fi adapter (\$15–\$25) is an inexpensive workaround while you arrange repair.

Section G — Troubleshooting Checklist

Done?

Restarted router and modem (unplugged for 30 seconds)



Ran netsh winsock reset, ip reset, ipconfig /release, /flushdns, /renew



Updated or reinstalled Wi-Fi driver in Device Manager



Checked physical Wi-Fi switch and Airplane Mode



Changed DNS to 8.8.8.8 / 8.8.4.4



Ran Windows Network Troubleshooter



Is This Your Problem?

Windows gets stuck in a loop. You see the spinning circle, then a black screen, then it restarts and does the same thing again. Or you see a message saying 'Automatic Repair', 'Recovery', or 'Preparing Automatic Repair' — and it never actually fixes anything.

Before starting: confirm your drive is physically detected (check Section F). If the drive isn't detected in BIOS, this section won't help — the drive needs to be replaced first.

Understanding Windows Recovery Options

Windows has built-in repair tools. We'll access them through the **Windows Recovery Environment (WinRE)** — a separate repair mode that loads even when Windows itself is broken.

Fix 1 — Access the Recovery Environment

1	Turn on the computer. When you see the Windows loading circle, press and hold the power button to force it off. Wait 5 seconds.
2	Turn it on again. Force it off again when you see the loading circle. Do this 3 times total.
3	On the 4th boot, Windows will automatically enter the Recovery Environment instead of trying to load normally.
4	You'll see a blue screen with options: 'Continue', 'Troubleshoot', or 'Turn off your PC'. Click Troubleshoot .
5	Click Advanced options .

Fix 2 — Startup Repair (Try This First)

1	From Advanced options, click Startup Repair .
2	Select your account (if asked) and enter your password.
3	Windows will scan for startup issues and attempt automatic repairs. This takes 5–20 minutes.
4	If it succeeds, the computer will restart normally into Windows.
5	If it fails with a message saying 'Startup Repair couldn't repair your PC' — continue to Fix 3.

Fix 3 — System Restore

If your computer was working and something changed (an update, an installation) that broke it, System Restore rolls Windows back to a previous state without affecting your personal files.

1	From Advanced options, click System Restore .
2	Select your account and enter your password.
3	Click Next. You'll see a list of restore points with dates.
4	Choose a restore point from before your computer started having problems.
5	Click Next, then Finish. Confirm when prompted.
6	The restore process takes 15–30 minutes. Do not interrupt it.
7	The computer will restart into Windows when complete.

Fix 4 — Repair Windows Files from Command Prompt

1	From Advanced options, click Command Prompt .
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2	In the black window, type: sfc /scannow and press Enter. Wait for completion.
3	Then type: DISM /Online /Cleanup-Image /RestoreHealth and press Enter. This can take 20+ minutes.
4	Type: chkdsk C: /f /r and press Enter. Type Y if prompted.
5	Type exit and press Enter. Choose to restart.

Fix 5 — Reset This PC (Keeps Your Files)

If repairs haven't worked, resetting Windows reinstalls the operating system while keeping your personal files (documents, photos, desktop items). Programs will need to be reinstalled afterward.

1	From Advanced options, click Reset this PC .
2	Choose Keep my files . Do NOT choose 'Remove everything' unless you intend to wipe the computer completely.
3	Choose either 'Cloud download' (downloads a fresh Windows — requires internet) or 'Local reinstall' (uses files already on your drive).
4	Follow the on-screen instructions. The process takes 30–90 minutes.
5	Your personal files will be intact. Reinstall your programs after the reset.

After a Windows reset, check Windows Update immediately and install all available updates. This ensures your system is secure and stable.

Fix 6 — Reinstall Windows from USB (Last Resort)

If the computer cannot even reach the Recovery Environment, you can create a bootable Windows USB from another computer and reinstall Windows from scratch. This will erase everything on the drive.

1	On a working computer, go to: microsoft.com/en-us/software-download/windows11 (or Windows 10).
2	Download the Media Creation Tool and run it.
3	Insert a USB drive (8GB or larger). Select 'Create installation media for another PC'.
4	Follow the wizard to create the bootable USB drive.
5	Insert the USB into the broken computer and start it up.
6	Enter BIOS (see Section F for BIOS key by brand) and look for 'Boot Order' or 'Boot Priority'.
7	Set the USB drive as the first boot device. Save and exit BIOS.
8	The computer will boot from USB and begin the Windows installation wizard.
9	Follow the prompts. You can choose to either keep files or do a clean install.

If this worked: Windows loads normally after Startup Repair, System Restore, or Reset. Run Windows Update and reinstall your programs.

If this didn't work: If the computer still won't boot after all fixes — including a USB reinstall — the drive has likely failed. See Section F and the pre-shop checklist on page 35.

Section H — Troubleshooting Checklist	Done?
Accessed Windows Recovery Environment using 3x forced-off method	☐
Ran Startup Repair — noted result	☐
Tried System Restore to a date before problems started	☐
Ran SFC, DISM, and CHKDSK from Command Prompt	☐
Used 'Reset this PC' with 'Keep my files' option	☐

Created bootable USB and attempted clean Windows reinstall (if all else failed)	
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Before You Call a Repair Shop

The complete pre-visit checklist

If you've worked through the relevant section in this guide and the problem persists, a repair shop is the right next step. But how you arrive at that appointment makes a significant difference in what you pay and what you get.

Technicians charge for diagnosis time. If you walk in having already ruled out the simple causes, they skip straight to the real problem — saving you 30–60 minutes of billable time on a typical visit.

Information to Have Ready

- **The exact error message or behavior** — write it down word for word, or photograph the screen
- **When it started** — did it happen suddenly, or gradually over weeks?
- **What changed before it started** — new update, new program, power outage, drop, spill?
- **Every troubleshooting step you've already tried** — use your section checklists
- **Your computer's age and model number** — usually on a sticker on the bottom or back
- **Whether the computer is still under warranty** — check the manufacturer's website with your serial number

What Not to Do Before Your Appointment

Do NOT attempt to reinstall Windows, format the drive, or do anything irreversible before you've backed up your data or confirmed with the shop what they'll need to see.

- Don't let a shop charge you for diagnosing something you've already diagnosed
- Don't agree to data recovery pricing without getting a second opinion — rates vary wildly
- Don't authorize repairs above your budget without a written quote first

- Don't hand over your computer without noting down your passwords and backing up what you can

Questions to Ask the Repair Shop

1. What is your diagnosis fee, and is it applied to the repair cost if I proceed?
2. Can you give me a written quote before doing any work?
3. What is your turnaround time, and do you have a loaner or rush option?
4. Is the repair cost-effective given my computer's age, or should I consider replacing it?
5. What warranty do you offer on the repair?
6. Will my data be backed up before you work on it?

Repair Cost Reference Guide

Use this table to calibrate whether a quote sounds reasonable:

Repair Type	Typical Cost Range	Notes
Virus/malware removal	\$50–\$120	Often DIY-able with Windows Security
RAM replacement	\$80–\$200+ depending on spec	RAM prices have risen sharply — DDR5 especially. Get the exact spec your board requires before buying.
SSD upgrade + data migration	\$80–\$180	Highly worthwhile on computers under 5 years old
OS reinstall (Windows)	\$60–\$120	DIY-able with this guide (Section H)
Power supply replacement	\$80–\$250+	A quality unit starts around \$80. Budget models should be avoided — a cheap PSU can damage other components.
Motherboard replacement	\$150–\$400+	Often not worth it on older PCs
Data recovery (failed drive)	\$100–\$1,500+	Cost varies widely by failure type; get quotes from multiple shops

Thermal paste replacement	\$50–\$100	Labor-intensive relative to the part cost; most shops charge \$50–\$100 for the full service.
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Preventive Maintenance Schedule

Keep your PC healthy year-round

The best repair is the one you never need. These habits, done consistently, prevent the majority of computer problems that this guide covers.

Weekly

- Restart your computer at least once (clears memory, installs pending updates)
- Check for and install Windows updates: Settings → Windows Update

Monthly

- Run Disk Cleanup (Windows key → 'Disk Cleanup') and empty the Recycle Bin
- Check storage: press Windows key → 'This PC'. Keep below 80% full
- Review Task Manager startup programs — remove anything new you don't need
- Back up important files to an external drive or cloud (Google Drive, OneDrive)

Every 6 Months

- Clean dust from all fans, heatsinks, and vents with compressed air
- Check that all case vents are unobstructed
- Update drivers: Device Manager → right-click major devices → Update driver
- Run a full Windows Security antivirus scan
- Test your backup: open a backed-up file and confirm it's readable

Annually

■	Check your surge protector — they lose effectiveness over time. Replace if 3+ years old
■	Consider thermal paste replacement if the computer is 4+ years old and running hot
■	Review whether any hardware upgrades (SSD, RAM) would meaningfully extend the computer's life
■	Check for firmware/BIOS updates from your motherboard manufacturer's website

Computers that receive consistent basic maintenance last significantly longer and rarely fail catastrophically. A 30-minute monthly routine can add years to the life of your machine.

Jargon Glossary

Every technical term in this guide — explained in plain English

You don't need to memorize any of this. Use this section as a reference whenever you encounter a term you're unsure about.

BIOS / UEFI	The first program your computer runs when it powers on — before Windows even starts. It checks that all hardware is present and then hands control to Windows. You can access it by pressing a specific key (F2, Delete, etc.) right after power-on. UEFI is the modern version of BIOS.
BSOD (Blue Screen of Death)	A Windows crash screen — blue with white text — that appears when something goes critically wrong. Contains an error code that identifies what failed. The computer restarts automatically after a BSOD.
Boot	The process of starting a computer. 'Booting up' means the computer is loading Windows. 'Boot device' refers to the drive that contains Windows.
CHKDSK	A Windows tool that scans your hard drive for errors and repairs them. Stands for 'Check Disk'. Run from Command Prompt with the command: <code>chkdsk C: /f /r</code>
CPU (Central Processing Unit)	The main processor — the 'brain' of the computer. Handles all calculations and instructions. Overheating CPUs throttle themselves (slow down) to prevent damage.
DISM	A Windows repair tool that can fix the Windows installation files themselves. More thorough than SFC. Run from Command Prompt: <code>DISM /Online /Cleanup-Image /RestoreHealth</code>
DNS (Domain Name System)	The system that translates website names (like google.com) into numerical addresses that computers understand. If DNS isn't working, you can be 'connected' to Wi-Fi but unable to load websites.
Driver	A small program that allows Windows to communicate with a hardware component (graphics card, Wi-Fi adapter, printer). Outdated or corrupted drivers are a very common cause of crashes and blue screens.
GPU / Graphics Card	The component responsible for producing the image on your screen. A dedicated graphics card is a separate board with its own memory and cooling fan. Some computers use graphics built into the CPU instead.

HDD (Hard Disk Drive)	A traditional storage device that uses spinning metal disks to store data. Slower than an SSD, makes audible sounds during operation, and is more vulnerable to physical damage. Being phased out in favor of SSDs.
Heatsink	A metal component (usually with fins) that sits on top of the CPU and absorbs heat. A fan blows air through the fins to cool the heatsink. Dust buildup here is a leading cause of overheating.
IP Address	A numerical address assigned to your computer on a network. When IP configuration gets corrupted, you can be connected to Wi-Fi but unable to reach the internet.
Motherboard	The large flat circuit board that everything else connects to — CPU, RAM, storage, graphics card, and more. Also called the 'main board'. A failed motherboard is typically the most expensive repair.
POST (Power-On Self Test)	The quick hardware check your computer performs every time it turns on, before loading Windows. A failed POST often produces beep codes.
PSU (Power Supply Unit)	The component that converts wall power into the voltages your computer components need. A failing PSU can cause random shutdowns, failure to turn on, or erratic behavior.
RAM (Random Access Memory)	Short-term memory your computer uses to run programs. Unlike storage, RAM is erased when the computer turns off. More RAM = ability to run more programs simultaneously without slowdown. Physically a flat stick that plugs into the motherboard.
SATA Cable	The thin, flat cable (usually red) that connects a storage drive to the motherboard. SATA cables can work loose over time, causing 'no boot device' errors.
SFC (System File Checker)	A Windows repair tool that scans for and replaces corrupted Windows system files. Run from Command Prompt: <code>sfc /scannow</code>
SSD (Solid State Drive)	A modern storage device with no moving parts. Much faster than an HDD, completely silent, and more resistant to physical shock. The recommended upgrade for any computer with a traditional hard drive.
System Restore	A Windows feature that saves 'restore points' — snapshots of your computer's settings at a point in time. Rolling back to a restore point can undo changes caused by a bad update or installation, without affecting personal files.

Thermal Paste

A heat-conducting compound applied between the CPU and its heatsink. Ensures efficient heat transfer. Dries out over 4–7 years and needs replacement to maintain proper CPU cooling.

Windows Recovery Environment (WinRE)

A separate repair mode built into Windows that loads when Windows itself can't start. Gives access to Startup Repair, System Restore, Command Prompt, and other tools. Accessed by interrupting the boot process 3 times.

You Now Know More Than Most.

Whether you've already fixed the problem yourself or you're walking into a repair shop fully prepared — you've done something most people never bother to do: you took control of the situation.

Computers are machines. Machines fail in predictable ways. Now you know what those ways are — and what to do about them.

Fix Any PC Problem Yourself — Beginner Edition

Save this guide somewhere you'll find it. Share it with anyone who's ever paid too much for a repair they didn't need.